

The Discriminant-12 Return

Centered Cayley Reciprocity, Pell–Lorentz Geometry, Cyclotomic Closure, and Finite Reduction

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2026

Abstract

For $g \in SL_2$ with trace τ , centered part $H = g - \frac{\tau}{2}I$, and Cayley transform $K = (g - I)(g + I)^{-1}$, we prove

$$K = \frac{2}{\tau + 2}H, \quad KH = \frac{\tau - 2}{2}I.$$

Thus, in the normalization $\tau = \sigma + 2$, the hyperbolic self-reciprocity condition $K = H^{-1}$ selects $\sigma = 2$, hence trace 4. With the period-[1, 2] positive integral normalization this yields

$$g_{12} = \begin{pmatrix} 3 & 1 \\ 2 & 1 \end{pmatrix}, \quad \lambda = 2 + \sqrt{3},$$

with characteristic discriminant 12. We realize g_{12} as multiplication by λ on the ramified ideal above 2 in $\mathbb{Q}(\sqrt{3})$, and its centered generator as multiplication by $\sqrt{3}$. The same unit has a separate factor-cone realization in null coordinates by $(x, y) \mapsto (\lambda x, \lambda^{-1}y)$ and translates rapidity by $\log \lambda$ on every real shell $xy = n$. The row/column parabola–circle tangent endpoints are its null eigenrays. After adjoining i , the operator $\mathcal{Z} = (H_{12} + iI)/2$ is multiplication by ζ_{12} on the integral cyclotomic ideal $\mathfrak{p}_2\mathcal{O}_{\mathbb{Q}(\zeta_{12})}$. At the ramified prime the cyclotomic residue field is $\mathbb{F}_4$: reduced multiplication by ζ_{12} has order 3, while the mod-2 coordinate reduction of the Pell return on the ramified ideal, transported through a compatible additive $\mathbb{F}_2$ -linear identification with $\mathbb{F}_4$, is Frobenius. Under the same identification, the reduced quadratic parity is exactly the finite-field trace, so Boolean XOR parity is $\text{Tr}_{\mathbb{F}_4/\mathbb{F}_2}$. Finally $n = 11$ appears as a distinguished specialization whose Artin action is T_{11} , complex conjugation, corresponding to the involution J in the companion representation. No statement about zeta zeros or the Riemann hypothesis is used.

1 Introduction

The construction begins before any arithmetic normalization. Centered Cayley reciprocity selects trace 4. A continued-fraction normalization then chooses a distinguished primitive positive integral representative. Its expanding eigenvalue is the fundamental Pell unit $2 + \sqrt{3}$, in the standard quadratic-unit setting of continued fractions and Pell equations [1, 2]. The same return is subsequently realized on an ideal lattice, as a Lorentz boost on the real factor cone, as a cyclotomic operator after adjoining i , and through compatible order-2 and order-3 finite actions after ramified reduction.

The point of the paper is the compatibility of these realizations, rather than any one of the standard background ingredients in isolation. In particular, the integral basis change on the ramified ideal reduces to the Boolean basis change, while the mod-2 Pell matrix becomes Frobenius after transport to the additive carrier of $\mathbb{F}_4$. This is distinct from literal residue-field multiplication by the Pell unit, which is trivial at the cyclotomic prime. The compatible additive identification also

makes the reduced quadratic parity an exact finite-field trace. The resulting chain is rigid enough to distinguish which identifications are algebraic and which are representation correspondences.

The logical chain is

$$\begin{aligned} K = H^{-1} &\iff \sigma = 2 \implies \tau = 4 \implies \Delta = 12 \\ &\implies \lambda = 2 + \sqrt{3} \implies g_{12} = \times \lambda \text{ on } \mathfrak{p}_2 \implies H_{12} = \times \sqrt{3} \\ &\implies \mathcal{Z} = \times \zeta_{12} \text{ on } \mathfrak{P}_2 \implies \mathbb{F}_4 \text{ residue and transported finite actions.} \end{aligned}$$

The real Cone is a parallel carrier of the same Pell return, not a finite quotient. Likewise the Boolean translation group, factor exchange, $U(12) \cong V_4$, and the affine/semilinear groups of $\mathbb{F}_4$ are distinct symmetry layers. Standard background on quadratic forms and ideal classes is used in the usual sense [3, 4]; cyclotomic and finite-field facts are used in their standard forms [5, 6].

The proof is organized in the same dependency order. We first establish the centered Cayley identity, then normalize the selected trace by the period $[1, 2]$ return, identify the ramified ideal and Lorentz norm, pass separately to the real factor cone, complete cyclotomically, and only then compare the four-state reductions. The $n = 11$ shell is treated last as a specialization rather than as an input.

2 Main theorem

Theorem 2.1 (Discriminant-12 return package). *Let $g \in SL_2(\mathbb{Q})$ have trace τ , assume $g + I$ is invertible, and set*

$$H = g - \frac{\tau}{2}I, \quad K = (g - I)(g + I)^{-1}, \quad \tau = \sigma + 2.$$

Then

$$K = \frac{2}{\tau + 2}H, \quad KH = \frac{\tau - 2}{2}I.$$

If g is hyperbolic, then

$$K = H^{-1} \iff \sigma = 2.$$

At this selected trace $\tau = 4$, impose the period- $[1, 2]$ positive integral normalization and define

$$g_{12} = \begin{pmatrix} 3 & 1 \\ 2 & 1 \end{pmatrix}, \quad H_{12} = g_{12} - 2I.$$

Then the following hold.

- (i) $\chi_{g_{12}}(X) = X^2 - 4X + 1$, $\Delta_{g_{12}} = 12$, and $H_{12}^2 = 3I$.
- (ii) On $\mathfrak{p}_2 = (2, \sqrt{3} - 1) = (1 + \sqrt{3})$, g_{12} and H_{12} are multiplication by $2 + \sqrt{3}$ and $\sqrt{3}$ respectively.
- (iii) The primitive form

$$q_{12}(m, n) = 2m^2 - 2mn - n^2$$

is preserved by g_{12} on ramified-ideal coordinates and satisfies

$$2q_{12} = U^2 - V^2, \quad (U, V) = (2m - n, \sqrt{3}n).$$

(iv) In the separate factor-cone realization,

$$x' = (2 + \sqrt{3})x, \quad y' = (2 - \sqrt{3})y,$$

so xy is invariant and rapidity is translated by $\log(2 + \sqrt{3})$. The factor-parabola tangent endpoints are the two null eigenrays.

(v) If $K_{12} = \mathbb{Q}(\zeta_{12})$ and $\mathfrak{P}_2 = \mathfrak{p}_2 \mathcal{O}_{K_{12}}$, then

$$\mathcal{Z} = \frac{H_{12} + iI}{2} = \times \zeta_{12}$$

on $\mathfrak{P}_2$, and $[\mathcal{O}_{K_{12}} : \mathbb{Z}[\sqrt{3}, i]] = 4$.

(vi) $\mathcal{O}_{K_{12}}/\mathfrak{P}_2 \cong \mathbb{F}_4$, and reduced $\mathcal{Z}$ is multiplication by an element ω of order 3. Separately, the mod-2 coordinate reduction of g_{12} on $\mathfrak{p}_2/2\mathfrak{p}_2$, transported through the compatible additive identification $f_1 \mapsto 1$, $f_2 \mapsto \omega^2$, is Frobenius on $\mathbb{F}_4$. These two $\mathbb{F}_2$ -linear actions generate $GL_2(\mathbb{F}_2) \cong S_3$.

3 Centered Cayley reciprocity

Lemma 3.1. For $g \in SL_2$ with trace τ ,

$$H^2 = \left(\frac{\tau^2}{4} - 1 \right) I.$$

Proof. Cayley–Hamilton gives $g^2 - \tau g + I = 0$. Substituting $H = g - \tau I/2$ gives the result. $\square$

Proposition 3.2. If $g + I$ is invertible, then

$$K = \frac{2}{\tau + 2} H, \quad KH = \frac{\tau - 2}{2} I.$$

Proof. Cayley–Hamilton gives

$$(g + I)((\tau + 1)I - g) = (\tau + 2)I,$$

whence

$$(g + I)^{-1} = \frac{(\tau + 1)I - g}{\tau + 2}.$$

Therefore

$$K = \frac{2g - \tau I}{\tau + 2} = \frac{2}{\tau + 2} H,$$

and multiplication by H gives the second identity. $\square$

Corollary 3.3. If g is hyperbolic and $\tau = \sigma + 2$, then

$$K = H^{-1} \iff \sigma = 2.$$

Proof. Hyperbolicity makes H invertible, so $K = H^{-1}$ iff $KH = I$, which is $(\tau - 2)/2 = 1$. $\square$

4 The primitive integral return and the ramified ideal

Let

$$A_a = \begin{pmatrix} a & 1 \\ 1 & 0 \end{pmatrix}.$$

The period $[1, 2]$ gives

$$g_{12} = A_1 A_2 = \begin{pmatrix} 3 & 1 \\ 2 & 1 \end{pmatrix}.$$

The Cayley condition selects the trace-4 hyperbolic class; the period/positivity normalization selects this representative. Directly,

$$\det g_{12} = 1, \quad \text{tr } g_{12} = 4, \quad \chi_{g_{12}}(X) = X^2 - 4X + 1,$$

so the eigenvalues are $2 \pm \sqrt{3}$ and the characteristic discriminant is 12. For

$$H_{12} = g_{12} - 2I = \begin{pmatrix} 1 & 1 \\ 2 & -1 \end{pmatrix},$$

one has $H_{12}^2 = 3I$.

Let

$$f_1 = 2, \quad f_2 = \sqrt{3} - 1, \quad \mathfrak{p}_2 = (f_1, f_2).$$

Then

$$\sqrt{3}f_1 = f_1 + 2f_2, \quad \sqrt{3}f_2 = f_1 - f_2,$$

so

$$[\times \sqrt{3}]_f = H_{12}, \quad [\times (2 + \sqrt{3})]_f = g_{12}.$$

This is a direct ideal-lattice realization in the quadratic field; no classification theorem is needed for the matrix computation itself.

A second natural basis is

$$e_1 = 1 + \sqrt{3}, \quad e_2 = 3 + \sqrt{3} = \sqrt{3}e_1.$$

Since $e_1 = f_1 + f_2$ and $e_2 = 2f_1 + f_2$,

$$C_{\text{basis}} = \begin{pmatrix} 1 & 2 \\ 1 & 1 \end{pmatrix}, \quad C_{\text{basis}}^{-1} g_{12} C_{\text{basis}} = \begin{pmatrix} 2 & 3 \\ 1 & 2 \end{pmatrix}.$$

Modulo 2, this integral basis change will become the Boolean relabeling.

5 Pell–Lorentz and factor-cone realizations

For ideal-lattice coefficients $m, n \in \mathbb{Z}$, write

$$\alpha = mf_1 + nf_2 = (2m - n) + n\sqrt{3}.$$

Then

$$N_{F/\mathbb{Q}}(\alpha) = (2m - n)^2 - 3n^2 = 2q_{12}(m, n),$$

where

$$q_{12}(m, n) = 2m^2 - 2mn - n^2.$$

Thus, with

$$U = 2m - n, \quad V = \sqrt{3}n, \\ 2q_{12} = U^2 - V^2.$$

Multiplication by $2 + \sqrt{3}$ is the Lorentz boost

$$\begin{pmatrix} U' \\ V' \end{pmatrix} = \begin{pmatrix} 2 & \sqrt{3} \\ \sqrt{3} & 2 \end{pmatrix} \begin{pmatrix} U \\ V \end{pmatrix}.$$

Now pass to the separate Paper-A factor coordinates $x, y > 0$ and define the full two-sided Cone by

$$X = \frac{x - y}{2}, \quad T = \frac{x + y}{2}, \quad Y^2 = xy, \quad Y = \pm\sqrt{xy}.$$

Then

$$T^2 - X^2 - Y^2 = 0, \quad x = T + X, \quad y = T - X.$$

The discriminant-12 boost has the factor-coordinate realization

$$\boxed{x' = \lambda x, \quad y' = \lambda^{-1} y,} \quad \lambda = 2 + \sqrt{3},$$

and $xy = n$ is invariant. If

$$x = \sqrt{n}e^s, \quad y = \sqrt{n}e^{-s},$$

then

$$\boxed{s' = s + \log(2 + \sqrt{3}).}$$

Factor exchange sends $X \mapsto -X$ and $s \mapsto -s$, while the independent cone-side reflection sends $Y \mapsto -Y$. The latter does not exchange the factors.

For fixed factor level $u > 0$, the Paper-A row $x = u$ is

$$Y^2 = u^2 - 2uX$$

with vertex $(u/2, 0)$, while the column $y = u$ is

$$Y^2 = u^2 + 2uX$$

with vertex $(-u/2, 0)$. A fixed- T Cone slice is

$$X^2 + Y^2 = T^2.$$

Tangency at the vertices occurs exactly for

$$T = u/2.$$

Hence the side-view tangent points are

$$(T, X) = \left(\frac{u}{2}, \pm\frac{u}{2}\right),$$

which are precisely the two null eigenrays of the discriminant-12 boost. Figure 1 displays the complete half-step fixed- T family through $x + y = 12$ and emphasizes the exact $u = 5, 6, 7$ tangencies. The figure is illustrative of the theorem above; the null-eigenray statement does not depend on the plotted discretization.

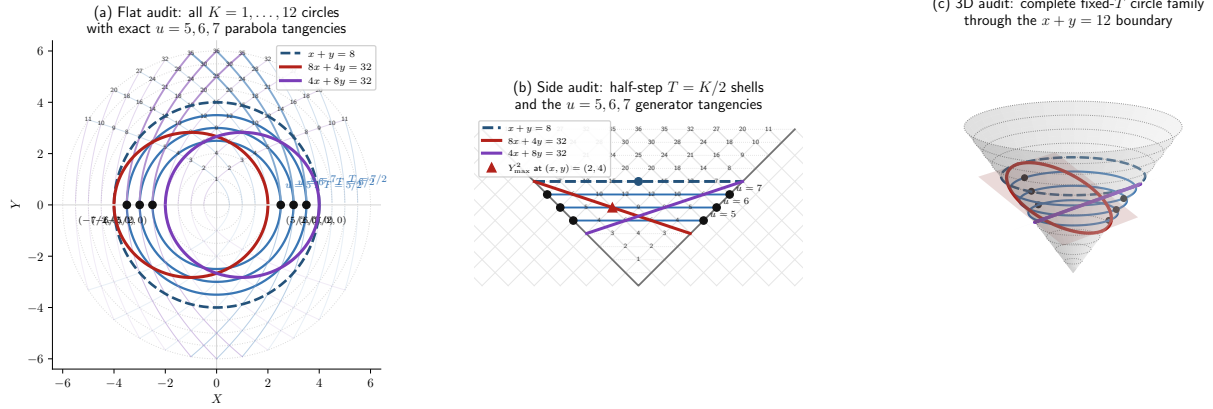


Figure 1: Parabola–circle tangency and the discriminant-12 null eigenrays. The flat projection, side view, and full two-sided Cone show the fixed- T family $T = K/2$ for $K = 1, \dots, 12$. The emphasized $u = 5, 6, 7$ row/column parabolas are tangent at $(T, X) = (u/2, \pm u/2)$, precisely the two null eigendirections of the Pell boost.

6 The mod-12 unit shell

The units modulo 12 are

$$U(12) = \{1, 5, 7, 11\} \cong V_4.$$

For $r \in U(12)$, the factor pairs $(r, 1)$ and $(1, r)$ have

$$T_r = \frac{r+1}{2}, \quad X_r = \pm \frac{r-1}{2}, \quad Y_r = \pm \sqrt{r}.$$

Thus the four labels occupy $T = 1, 3, 4, 6$, and the outer $r = 11$ pair lies on $x + y = 12$. The standard cyclotomic action is

$$\text{Gal}(\mathbb{Q}(\zeta_{12})/\mathbb{Q}) \cong U(12), \quad T_r : \zeta_{12} \mapsto \zeta_{12}^r,$$

as usual for cyclotomic fields [5]. The geometry carries the labels but does not define their modular group law.

7 Cyclotomic completion on an integral ideal

Let

$$K_{12} = \mathbb{Q}(\zeta_{12}) = \mathbb{Q}(\sqrt{3}, i), \quad \zeta = \frac{\sqrt{3} + i}{2}.$$

Since $H_{12} = \times \sqrt{3}$,

$$\mathcal{Z} = \frac{H_{12} + iI}{2} = \times \zeta.$$

The correct integral carrier is

$$\mathfrak{P}_2 = \mathfrak{p}_2 \mathcal{O}_{K_{12}}.$$

For $\alpha = 1 + \sqrt{3}$,

$$\{\alpha, \alpha\zeta, \alpha\zeta^2, \alpha\zeta^3\}$$

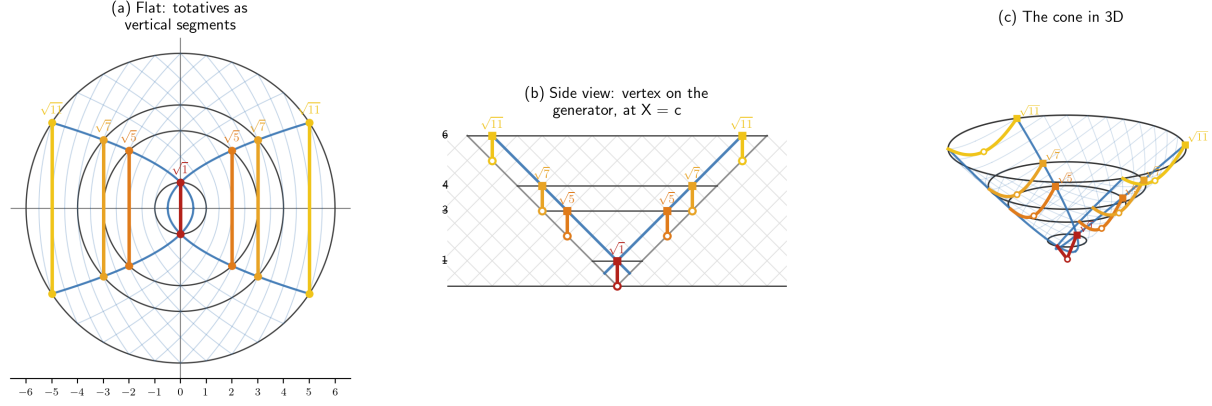


Figure 2: The mod-12 unit labels in the two-sided factor-cone coordinates. For each factor pair the full Cone contains the reflected lifts $Y = \pm\sqrt{r}$. The outer $r = 11$ pair is $(11, 1)$ and $(1, 11)$ on $x + y = 12$. The geometric placement realizes the four labels but is not itself the multiplication law of $U(12)$.

is a $\mathbb{Z}$ -basis of $\mathfrak{P}_2$. Since

$$\Phi_{12}(X) = X^4 - X^2 + 1, \quad \zeta^4 = \zeta^2 - 1,$$

multiplication by ζ has matrix

$$\begin{pmatrix} 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}.$$

Thus the half in $(H_{12} + iI)/2$ is a coordinate effect, not a failure of integrality on the correct lattice.

The full ring of integers is $\mathcal{O}_{K_{12}} = \mathbb{Z}[\zeta_{12}]$, whereas $\mathcal{O}_0 = \mathbb{Z}[\sqrt{3}, i]$ is a suborder of index

$$[\mathcal{O}_{K_{12}} : \mathcal{O}_0] = 4.$$

Equivalently,

$$\text{disc}(\mathcal{O}_0) = 2304, \quad \text{disc}(\mathcal{O}_{K_{12}}) = 144.$$

These are standard algebraic-number-theoretic notions of integral closure and field discriminant [4, 5]. The additive quotient $\mathcal{O}_{K_{12}}/\mathcal{O}_0 \cong (\mathbb{F}_2)^2$ records integral-closure gluing; it is not the residue field below.

8 Ramified reduction, transported Frobenius, and XOR trace

Modulo 2 in the ramified basis,

$$\bar{g}_{12} = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \quad \bar{q}_{12}(m, n) = n.$$

Also

$$C_{\text{basis}} \equiv \Phi = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix} \pmod{2},$$

and therefore

$$\Phi \bar{g}_{12} \Phi^{-1} = P = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

Thus the Boolean relabeling is the reduction of an integral ideal basis change. In Boolean coordinates (ϵ_1, ϵ_2) ,

$$\bar{q}_{12} = \epsilon_1 \oplus \epsilon_2.$$

Together with Boolean translations, factor exchange gives $V_4^{\text{tr}} \rtimes C_2 \cong D_8$.

The cyclotomic residue field is a different four-state ring:

$$\boxed{\mathcal{O}_{K_{12}}/\mathfrak{P}_2 \cong \mathbb{F}_4.}$$

Let $\omega = \bar{\zeta}$. Then

$$\omega^2 + \omega + 1 = 0, \quad \omega^3 = 1.$$

In the additive basis $(1, \omega)$,

$$[\times \omega] = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}, \quad [\text{Fr}] = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}.$$

The matrix equality with $\bar{g}_{12}$ is obtained only after the compatible additive identification described below. It is not literal residue-field multiplication by the Pell unit. Indeed, in $\mathcal{O}_{K_{12}}/\mathfrak{P}_2$,

$$2 \equiv 0, \quad \sqrt{3} \equiv -1 \equiv 1, \quad \lambda = 2 + \sqrt{3} \equiv 1,$$

so direct residue multiplication by λ is the identity.

By contrast, multiplication by ζ_{12} reduces literally to multiplication by ω . After transporting the mod-2 Pell matrix to the same additive carrier, one has

$$\boxed{\psi \bar{g}_{12} \psi^{-1} = \text{Fr}}, \quad \text{Fr}(\times \omega) \text{Fr}^{-1} = (\times \omega)^{-1},$$

so

$$\langle \times \omega, \text{Fr} \rangle = GL_2(\mathbb{F}_2) \cong S_3.$$

With translations,

$$\text{AGL}(1, 4) \cong A_4, \quad \text{AFL}(1, 4) \cong S_4.$$

The Frobenius and trace statements here are the standard degree-2 finite-field constructions specialized to $\mathbb{F}_4$ [6].

8.1 Compatible additive identification and finite-field trace

Define the $\mathbb{F}_2$ -linear identification

$$\psi : \mathfrak{p}_2/2\mathfrak{p}_2 \longrightarrow \mathbb{F}_4$$

by

$$\boxed{f_1 \longmapsto 1, \quad f_2 \longmapsto \omega^2.}$$

Because $e_1 = f_1 + f_2$ and $e_2 = f_2$ modulo 2, this gives

$$e_1 \longmapsto 1 + \omega^2 = \omega, \quad e_2 \longmapsto \omega^2.$$

Thus the principal/Pell basis corresponds to (ω, ω^2) , a basis interchanged by Frobenius. Consequently the same transported involution is represented as the transvection $\bar{g}_{12}$ in the ramified basis and as factor exchange P in the principal/Boolean basis.

The field trace satisfies

$$\mathrm{Tr}_{\mathbb{F}_4/\mathbb{F}_2}(1) = 0, \quad \mathrm{Tr}_{\mathbb{F}_4/\mathbb{F}_2}(\omega) = \mathrm{Tr}_{\mathbb{F}_4/\mathbb{F}_2}(\omega^2) = 1.$$

Hence in the ramified basis

$$\mathrm{Tr}_{\mathbb{F}_4/\mathbb{F}_2}(m + n\omega^2) = n = \bar{q}_{12}(m, n),$$

while in the Boolean/Pell basis

$$\boxed{\mathrm{Tr}_{\mathbb{F}_4/\mathbb{F}_2}(\epsilon_1\omega + \epsilon_2\omega^2) = \epsilon_1 \oplus \epsilon_2.}$$

Therefore, under this compatible additive identification,

$$\boxed{\bar{q}_{12} = \mathrm{Tr}_{\mathbb{F}_4/\mathbb{F}_2},}$$

so Boolean XOR parity is exactly the finite-field trace.

Remark 8.1 (Carrier guardrail). The map ψ is an additive $\mathbb{F}_2$ -linear identification, not a ring isomorphism. The ramified quotient $\mathfrak{p}_2/2\mathfrak{p}_2$ is nonreduced, whereas $\mathcal{O}_{K_{12}}/\mathfrak{P}_2 \cong \mathbb{F}_4$ is a field. In particular, the transported Frobenius statement and literal residue multiplication by λ are different constructions.

9 The distinguished $n = 11$ specialization

The construction above does not require $n = 11$. It enters afterward as a distinguished specialization. Geometrically, the two factor pairs

$$(11, 1), (1, 11)$$

lie on $xy = 11$ and $x + y = 12$. On the full two-sided Cone they have the four lifts

$$X = \pm 5, \quad Y = \pm\sqrt{11}, \quad T = 6.$$

Figure 3 places this shell in the same three views used for the Cone geometry and overlays the corresponding divisor summatory quantities. Its purpose is explanatory: the arithmetic specialization below is established independently of the visualization.

Arithmetically,

$$N_{F/\mathbb{Q}}(1 + 2\sqrt{3}) = -11.$$

Let

$$\mathfrak{p}_{11} = (1 + 2\sqrt{3}).$$

Since

$$(1 + \sqrt{3})(1 + 2\sqrt{3}) = 7 + 3\sqrt{3}$$

is totally positive,

$$\boxed{[\mathfrak{p}_{11}] = [\mathfrak{p}_2]}$$

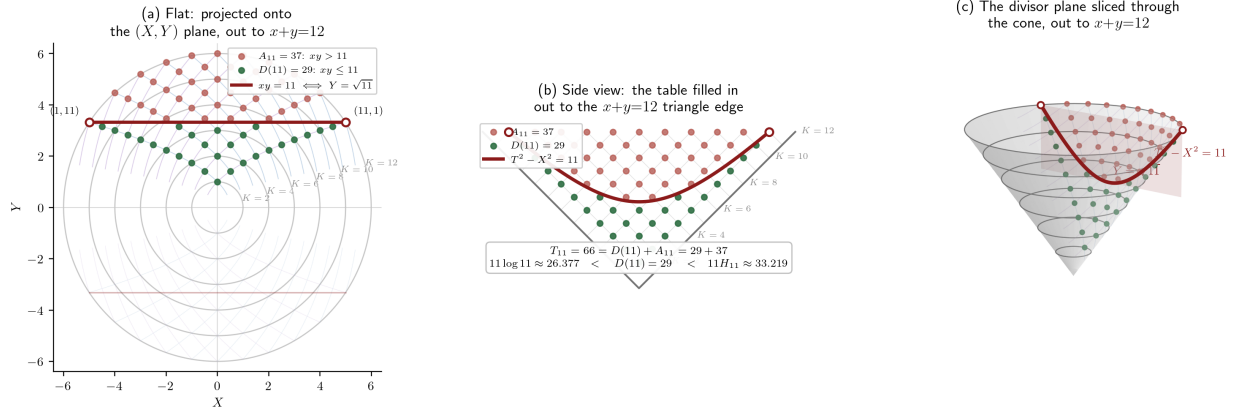


Figure 3: The $n = 11$ divisor shell in the flat, side, and three-dimensional Cone views. The full two-sided shell contains the reflected lifts $Y = \pm\sqrt{11}$ above the factor endpoints $X = \pm 5$, $T = 6$. The figure relates this shell to the divisor summatory count and the comparison quantities used in Paper A.

in the narrow ideal class group. For the narrow Hilbert class field $K_{12} = \mathbb{Q}(\zeta_{12})$ of $F = \mathbb{Q}(\sqrt{3})$,

$$\text{Art}_{K_{12}/F}(\mathfrak{p}_{11}) = T_{11}.$$

The use of the narrow class group and its Artin reciprocity map is standard class-field theory terminology [4, 3]. Here T_{11} denotes the nontrivial element of the quadratic subgroup $\text{Gal}(K_{12}/F) \subset \text{Gal}(K_{12}/\mathbb{Q})$ obtained from the cyclotomic automorphism with exponent 11. It lies in this subgroup because

$$T_{11}(\sqrt{3}) = \sqrt{3},$$

and, since $11 \equiv -1 \pmod{12}$,

$$T_{11}(\zeta_{12}) = \zeta_{12}^{11} = \zeta_{12}^{-1} = \overline{\zeta_{12}}.$$

Thus T_{11} is complex conjugation over F .

In the companion operator representation this involution corresponds to J :

$$T_{11} \longleftrightarrow J.$$

This is a representation correspondence, not a literal equality between a field automorphism and an operator in another category. The specialization chain is

$$11 \longrightarrow \mathfrak{p}_{11} \longrightarrow [\mathfrak{p}_{11}] = [\mathfrak{p}_2] \longrightarrow T_{11} \longleftrightarrow J.$$

10 Symmetry layers and scope

The finite groups appearing above have different roles:

$ \begin{array}{l} U(12) \cong V_4 \\ V_4^{\text{tr}} \\ V_4^{\text{tr}} \rtimes C_2 \cong D_8 \\ \mathbb{F}_4^\times \cong C_3 \\ \text{AGL}(1, 4) \cong A_4 \\ \text{AGL}(1, 4) \cong S_4 \end{array} $	$ \left \begin{array}{l} \text{cyclotomic Galois labels,} \\ \text{Boolean translations,} \\ \text{translations plus factor exchange,} \\ \text{nonzero residue-field multiplication,} \\ \text{affine field symmetry,} \\ \text{affine semilinear symmetry.} \end{array} \right. $
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They are compatible layers, not interchangeable groups.

A separate analytic companion may realize the same Pell unit through a certified operator crossing. That realization is not used here. In particular, the present paper does not infer analytic zero locations from the finite reductions, the class-field description, or the Cone geometry. It proves no statement about zeta-zero locations and makes no claim concerning the Riemann hypothesis.

11 Conclusion

Centered Cayley reciprocity selects trace 4 before arithmetic data enter. The positive period-[1, 2] normalization then exposes the Pell unit $2 + \sqrt{3}$ and fixes the integral representative g_{12} . The subsequent steps do not identify unlike carriers; they give compatible realizations of the same discriminant-12 return. On the ramified ideal it is multiplication by the Pell unit; after centering it is multiplication by $\sqrt{3}$; on the real factor cone it has a Lorentz-boost realization with null eigenrays equal to the exact parabola–circle tangent endpoints; and after cyclotomic completion the centered operator becomes multiplication by ζ_{12} on an integral rank-4 ideal.

At the finite level, multiplication by ζ_{12} reduces literally to the order-3 field multiplication by ω . The order-2 Frobenius action is a different construction: it is the mod-2 Pell coordinate action on the ramified ideal transported to the additive carrier of $\mathbb{F}_4$. Literal residue multiplication by λ is the identity. Under the same compatible additive identification, the reduced quadratic parity is the field trace

$$\bar{q}_{12} = \text{Tr}_{\mathbb{F}_4/\mathbb{F}_2},$$

so Boolean XOR is not an added convention but the trace of the common four-state additive carrier. The $n = 11$ shell finally supplies a distinguished specialization in which the outer $x + y = 12$ geometry, the ramified narrow class, cyclotomic complex conjugation T_{11} , and the companion involution J meet without being used to define the theory.

The resulting package is therefore best read as an exact compatibility statement across real, integral, cyclotomic, and finite representations of a single discriminant-12 return, with the reduction maps and carrier changes kept explicit. Any further analytic application is a separate problem and requires additional hypotheses and proofs.

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